USER GUIDE

MTI Radio Management

The register of MTI's handheld (HT) and rig radios: the register itself, handover and the checkout form, work from the automatic checks, service, stock opname, the COF Scan and ICT Masterlist documents from SharePoint, and the IT assets of Snipe-IT.

Application version: 1.21.4 · **Date:** 21 October 2026

For: the users of the application: the ICT team, admins, viewers and document controllers.

The screenshots in this guide use sample data, not real radio data.

Contents

1. The application at a glance 3
2. Signing in and out 5
3. The screen and the Dashboard 7
4. Register Radio 12
5. Radio details, handover and the checkout form 15
6. Needs Attention 18
7. Service & Repair 20
8. Stock Opname 22
9. History, Master Data, E-Sign Master and Import Excel 24
10. COF Scan and ICT Masterlist 26
11. Changing the ICT Masterlist 28
12. IT Assets: Snipe-IT 31
13. Administration 34
14. Messages you may see 37

1. The application at a glance

MTI Radio Management keeps the register of every MTI handheld (HT) and rig radio: one row per physical radio, known by its SN, who holds it, and its history. It also turns the checks of the register into a list of work, follows radios that are in service, helps with stock opname from a phone, opens the COF Scan and ICT Masterlist documents from SharePoint, and shows the IT assets of Snipe-IT.

The sidebar follows the role of your account: menus you may not open are not shown, and neither are buttons you may not use. The table below is the access of the standard roles; Super Admin can change it in **Role & Permission**.

Sidebar section	Menu	Super Admin	Admin Radio	Viewer Radio	Document Controller
–	Dashboard	Full	Full	View	View
RADIO	Register Radio, Needs Attention, Service & Repair, Stock Opname, Radio History, Import Excel	Full	Full	View	–
RADIO	Radio Master Data: Radio Types, Companies, Departments, Channel Templates	Full	Full	View	–
RADIO	Radio Master Data: E-Sign Master	Full	–	–	–
DOCUMENTS	COF Scan	Full	View	–	View
DOCUMENTS	ICT Masterlist	View, edit a row directly	View, ask for changes	–	View, ask for changes, edit a row directly
DOCUMENTS	Edit Approvals	Approve / reject	View	–	Approve / reject
IT ASSETS	Snipe-IT: Overview, Assets, Catalog, People & Organization, Licenses & Stock, History	View	–	–	–
USER	User Management: Registered Users, Role & Permission	Full	–	–	–
SETTINGS	Audit Log, API Tokens	Full	–	–	–

Full means seeing, adding, changing and deleting; **View** means seeing only. A request to change a cell is never approved by the person who asked for it; those who may approve can change a row directly (chapter 11). The Snipe-IT menu only reads, whoever opens it (chapter 12).

The application can be shown in English or Indonesian (chapter 3). This guide uses the English names of the menus and buttons; in Indonesian they read differently, for example **Perlu Perhatian** for Needs Attention and **Persetujuan Edit** for Edit Approvals.

Words you will often see

- **Radio ID**: the number of the radio in the system, unique among the active radios.
- **MDC Code**: the Radio ID in hexadecimal (1001 → 3E9), worked out automatically. An ID whose MDC Code contains the letter F cannot be used.
- **SN**: the serial number, unique; one row of the register is one physical radio.

- **Location**: With user, IT stock, In service or Lost.
- **Condition**: Good, Needs service, Broken or Not checked.
- **Owner**: MTI or Non MTI, following the asset owner of the radio.
- **Checks**: automatic checks of every radio (duplicate SN, empty ID, verification over 12 months ago, and more); their results are worked on in **Needs Attention**.
- **Service ticket**: the record of one radio sent to a service center, numbered `SRV-0001`.
- **Snipe-IT**: the company's inventory of IT assets (laptops, monitors, CCTV, radios and more). This application only reads it; changes are still made in Snipe-IT.

2. Signing in and out

Sign in with the **Microsoft** button or with your **Merdeka Account**. An admin must have added your account in User Management and given it a role; there is no self-registration and no password reset.

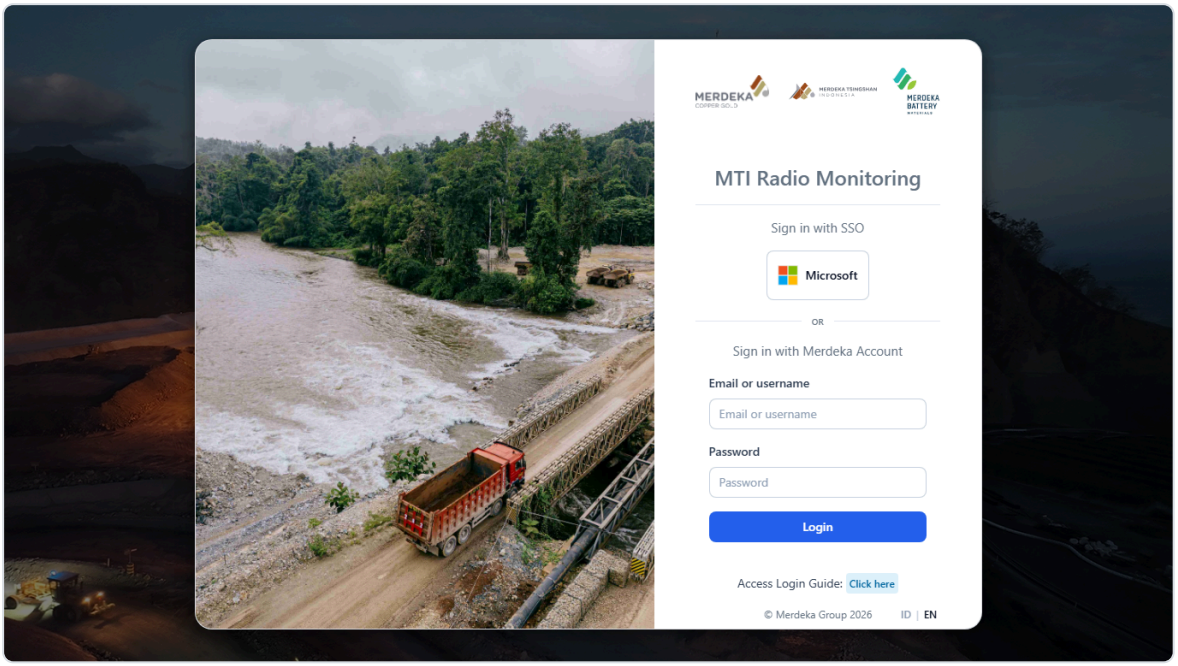


Figure 1. The login page: the Microsoft button at the top, the Merdeka Account form below it, and the language choice ID | EN at the bottom right.

Signing in with Microsoft

1. Open the address of the application, then click **Microsoft** under "Sign in with SSO".
2. Choose your Microsoft account (@merdekabattery.com). The choice of account always appears, so make sure you pick your own.
3. Finish the Microsoft sign-in. You land on the home page of your role.

Signing in with a Merdeka Account

1. Fill in **Email or username** with your e-mail address. If your account uses the LDAP method, your Windows username works too (for example budi.santoso, without @...).
2. Fill in **Password**: your Windows (Active Directory) password for an LDAP account, or the password an admin gave you.
3. Click **Login**. This form allows 10 attempts a minute.

The **Click here** button beside "Access Login Guide" shows how to sign in. The login page is in Indonesian at first; click **EN** at the bottom right for English, or **ID** to go back.

Session time limits

A session ends after 30 minutes without activity, or 8 hours after signing in even if you stay active. Activity means opening a page or using a button; the notification bell checking by itself does not count.

When	What appears	Choices
28 minutes without activity	"Your session is about to end" with a 2-minute countdown	**Stay signed in** keeps the session; **Sign out** signs out now

When	What appears	Choices
7 hours 55 minutes after signing in	"Your session is about to end" with a 5-minute countdown	**Sign in again now**, or **Later** to finish your work first
The limit is reached	The page moves to the login page with the message "Session expired" and the reason	Sign in again

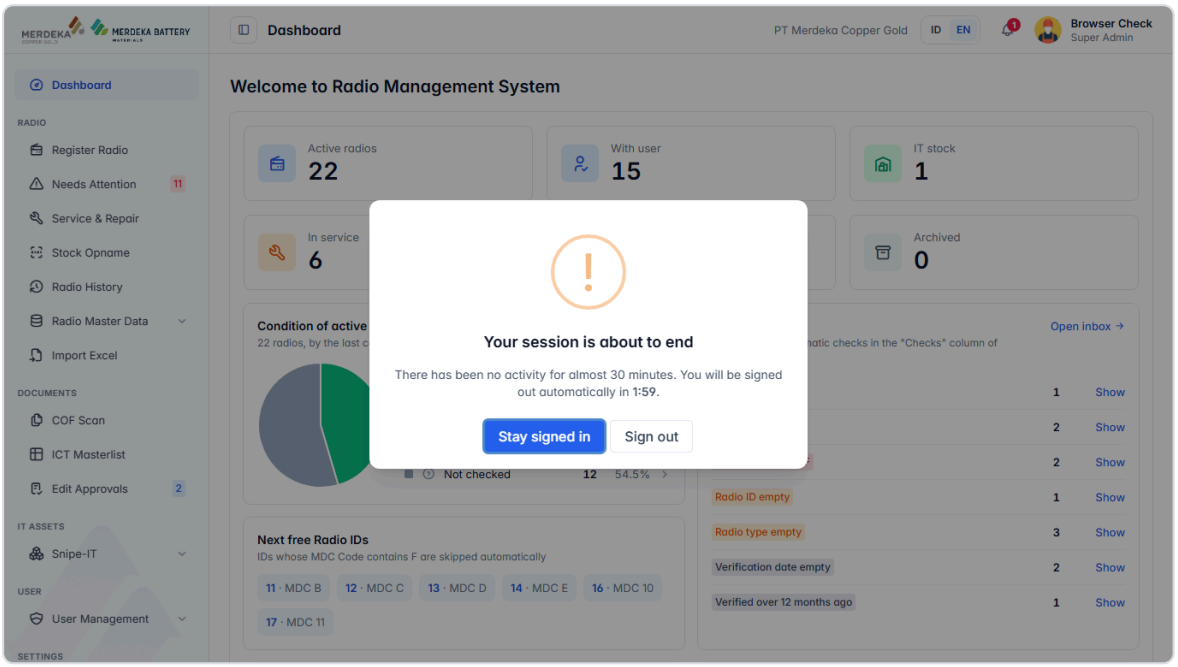


Figure 2. The session warning: a 2-minute countdown with the buttons Stay signed in and Sign out.

> Save your work before the countdown ends: what you typed in a form and did not save is lost when the session ends.

Signing out

Click your name at the top right, then **Sign out**. This signs you out of this application only; the Microsoft account in the browser stays signed in.

3. The screen and the Dashboard

Every page has the sidebar menu on the left, a top bar with the page title, the breadcrumb below it, the content of the page (usually one card with its tools and table), and a footer with the version of the application. The Dashboard is the home page that sums up the whole register.

Parts of the screen

- **Sidebar**: the menus by section (RADIO, DOCUMENTS, IT ASSETS, USER, SETTINGS). The red number on **Needs Attention** counts the open items, the blue number on **Edit Approvals** the requests waiting. The button at the left end of the top bar narrows the sidebar to a row of icons; point at an icon to see the name of its menu. The browser remembers this choice.
- **Top bar**: the page title, the name of your company, the language choice **ID | EN**, the notification bell, and your name and role. Click your name to open **Notifications** or **Sign out**.
- **Breadcrumb** below the top bar: where the page sits in the menu, for example "Dashboard › Register Radio › Radio 201". Click a part of it to go back to that page; the names of groups such as Radio Master Data are plain text. The Dashboard has no breadcrumb, and on a phone, which does not show the top bar, the page title is below the breadcrumb.
- **Footer**: the version of the application, for example v1.21.4. Click it to see the **Version history**, and mention the number when you report a problem.
- **Result messages**: after you save something, a short message appears for a moment at the bottom of the screen (for example "Ticket SRV-0012 created."). Confirmations before deleting or archiving appear in the middle of the screen.

Language

Click **ID** or **EN** in the top bar to show the application in Indonesian or English. On a phone this choice is in the sidebar. It is saved with your account, so it follows you to other devices; the default is Indonesian.

- What changes: menus, titles, buttons, tables, dialogs, confirmations, messages from the server, the names of months, and new notifications (written in the language of whoever gets them).
- The data stays as it is stored, in Indonesian: radio history, notes, the entries of the Audit Log, the content of SharePoint, Excel files, the printed checkout form, and the labels of the REST API. Locations, conditions, archive reasons, accessories and kinds of history are only shown by their names in the language of the page. In the screenshots of this guide you will see such Indonesian data, for example the descriptions under Recent activity.

Version history

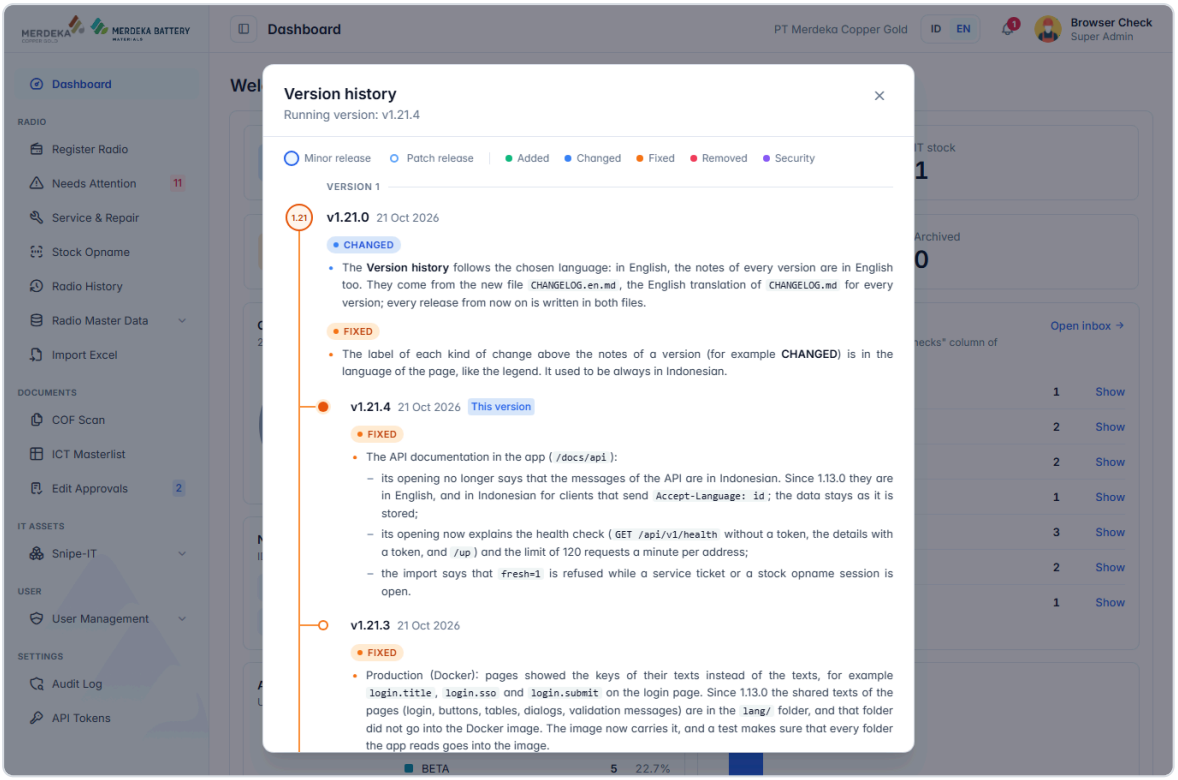


Figure 3. Version history: the legend, the running version marked This version, and the patch releases branching off the 1.21 series.

The version in the footer opens the **Version history** dialog: what changed in each version of the application, newest first. Every signed-in account can open it.

- Each series (for example 1.21) is one node of the timeline, and its patch releases (1.21.1, 1.21.2, …) branch off below it. The band **Version 1** marks where a major version begins.
- The colour of a version follows its strongest kind of change, and each part of its notes carries the label of its kind. The legend and these labels are always in English, in the terms of Semantic Versioning and Keep a Changelog: **Minor release**, **Patch release**, **Added**, **Changed**, **Fixed**, **Removed**, **Security**.
- The running version is marked **This version**. The notes follow the chosen language.

The notification bell

The red number on the bell counts the unread notifications and is updated every minute. Click the bell to see the latest ones; click one to mark it read and open its page. **Mark all as read** clears the number, and **See all notifications** opens the **Notifications** page.

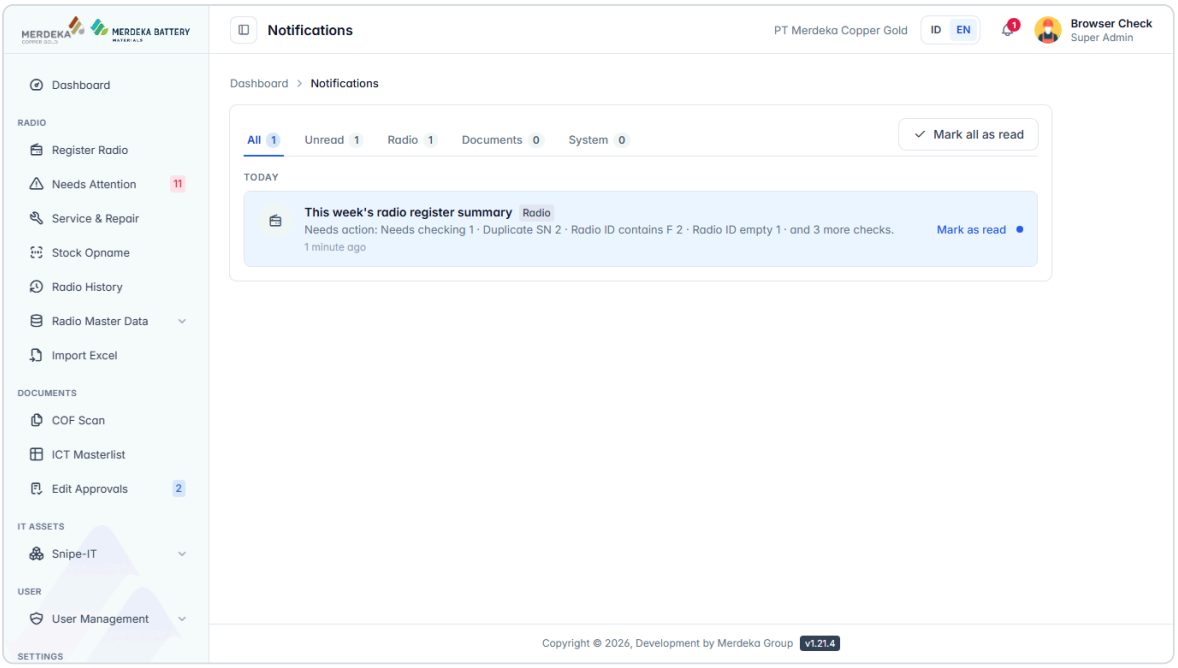


Figure 4. The Notifications page: tabs with their counts, and Mark as read on an unread row.

The Notifications page has the tabs **All**, **Unread**, **Radio**, **Documents** and **System**, each with its count. Notifications are grouped Today, Yesterday and Earlier; the time reads "12 minutes ago" for today, "Yesterday, 16:40", or "17 Oct, 08:00" for older ones, with the full date and time in a tooltip. **Mark as read** on a row marks it without opening the notification.

Notification	Tab	Who gets it
The weekly summary of the register ("Needs action")	Radio	Whoever may change Register Radio
A request to change an ICT Masterlist cell	Documents	Whoever may approve it, except the requester
Your request was decided: Applied, Rejected, Failed or Conflict	Documents	The requester
Your API token expires within 7 days	System	The owner of the token (an API client's token: Super Admin)

Notifications only appear in the application; they are not sent by e-mail. Read ones are deleted automatically after 90 days.

Dashboard

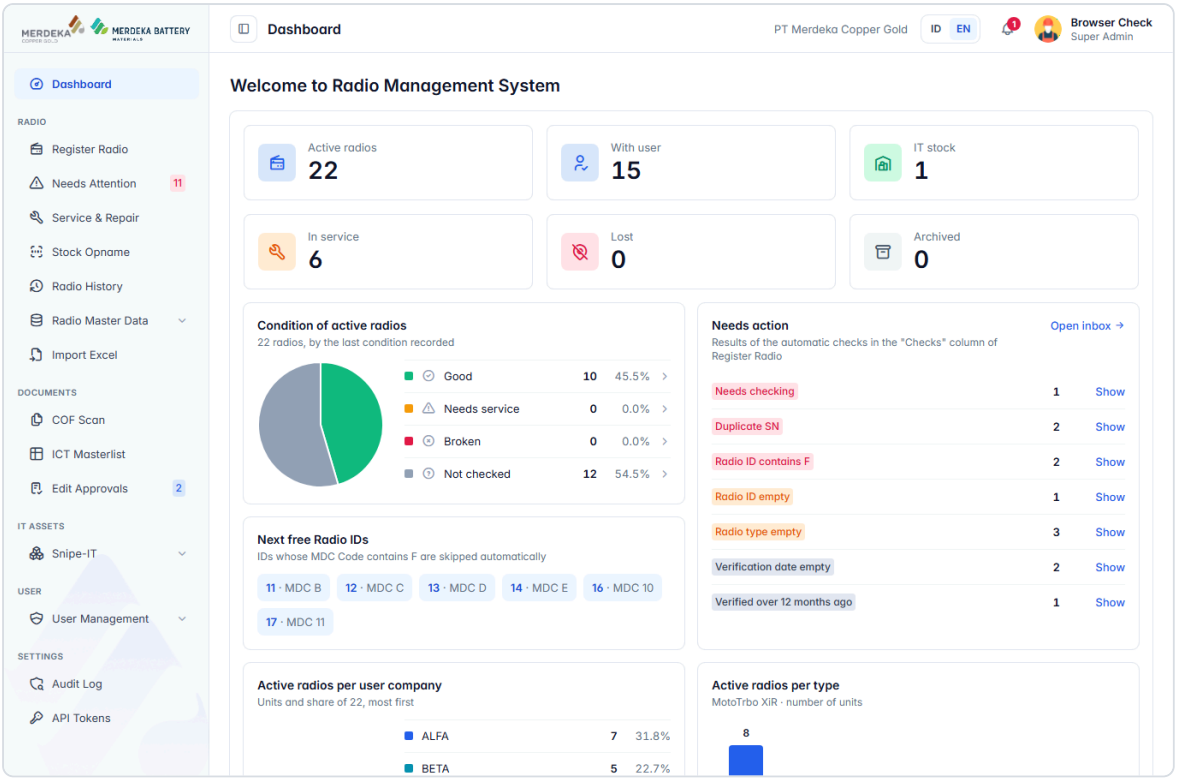


Figure 5. Dashboard: six number cards, the condition of active radios, the next free Radio IDs, and Needs action.

The Dashboard opens with the title "Welcome to Radio Management System", followed by these panels.

Panel	What it shows
Number cards	Active radios, the count per location (With user, IT stock, In service, Lost) and the archived ones. Click a card to open Register Radio already filtered
Condition of active radios	A pie chart of Good, Needs service, Broken and Not checked, with a list of counts and percentages. Click a slice or its row to open the list of those radios
Next free Radio IDs	IDs that a new radio can use; IDs whose MDC Code contains F are skipped automatically
Needs action	Problems found by the Checks column (for example Duplicate SN, Radio ID contains F, Verified over 12 months ago) with counts and a **Show** link; **Open inbox** opens Needs Attention
Active radios per user company	A ring with the number of radios in the middle and a list of counts and percentages. The five largest companies each have a colour of their own, radios without a company read "(none)", and the other companies are grouped as "Other" (their names written below it)
Active radios per type	A bar chart, with the count above each bar and the name of the type below it
Recent activity	The latest history (handover, service, verification and more), with links to the radios; **All history** opens Radio History

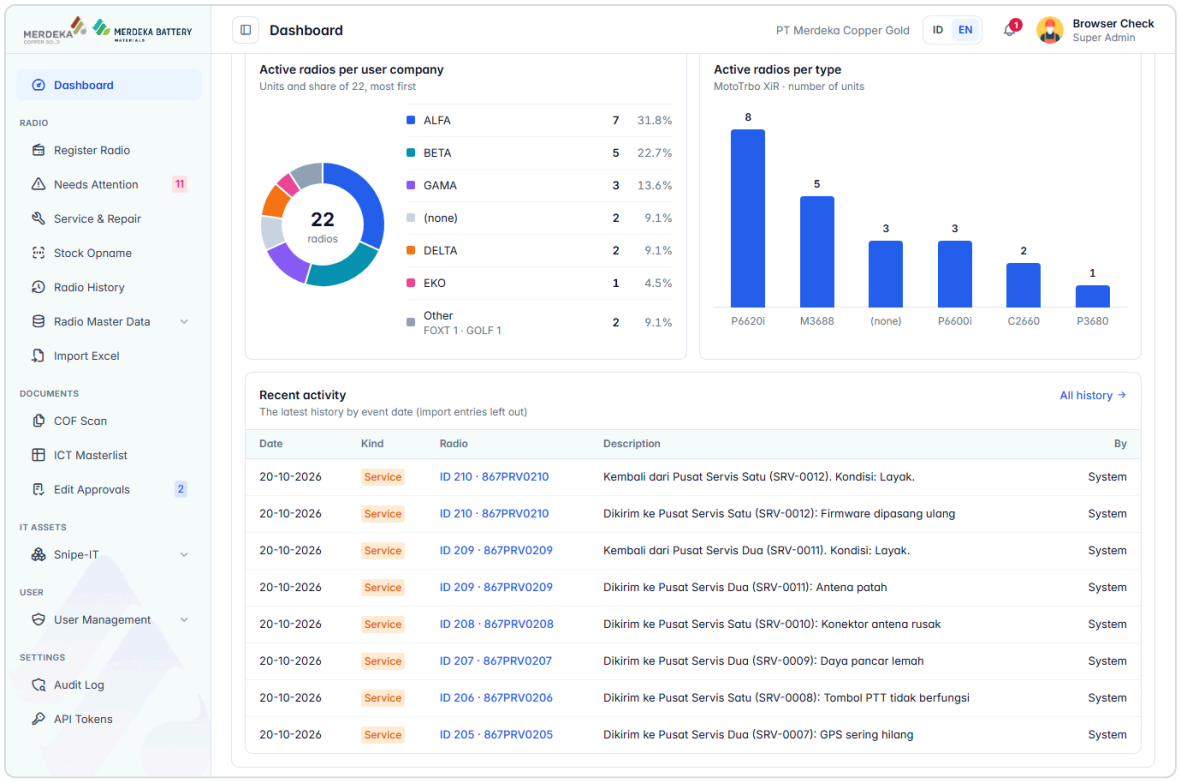


Figure 6. The lower part of the Dashboard: active radios per user company and per type, then Recent activity.

4. Register Radio

Register Radio is the list of all radios: one row for one physical radio, known by its SN. Here radios are searched, added, changed, archived, have their checkout form printed and are exported to Excel. Click a row to open the details of the radio.

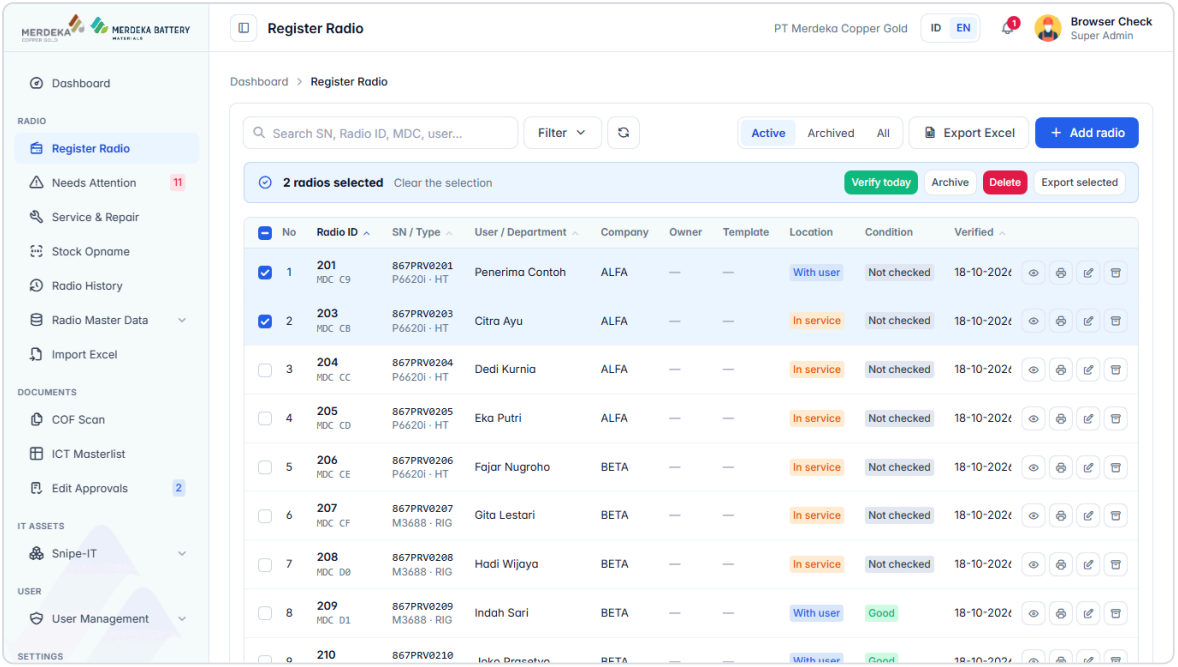


Figure 7. Register Radio with two radios ticked: the bar for many radios at once shows above the table.

Searching and filtering

- Type in the search box to search by SN, Radio ID, MDC Code or user name; the list changes as you type.
- Choose **Active**, **Archived** or **All** for active radios, archived radios, or both. The Archived view has its own columns: the last user, the archive date and the reason.
- Click **Filter** to filter by Company (user), Asset owner, Owner, Department, Radio type, Channel template, Location, Condition or Check result, then **Apply filters**. Active filters show as chips after the word **Applied**; click the × on a chip to remove it, or **Clear filters** in the Filter panel to remove them all.
- The **Refresh** button (the circular arrow beside Filter) fetches the latest list without clearing the search and the filters.
- Click the heading of the Radio ID, SN, User or Verified column to sort. Below the table, choose the **Rows per page** and move between pages.
- **Export Excel** exports every radio that matches the filters in use.

The **Checks** column marks problems of a radio: a duplicate SN or ID, an ID containing F, an empty SN, ID or type, lost without a BA, needs reprogramming, an empty verification date or one more than 12 months ago, and **Needs checking** (old Excel data that contradicts itself and was not guessed at). These problems are worked on in **Needs Attention** (chapter 6).

The buttons at the end of each row: **Details** (eye), **Print the form** (printer), **Edit** (pencil) and **Archive** (box). They stay visible when the table scrolls sideways.

Adding or changing a radio

1. Click **Add radio**, or the pencil (**Edit**) on the row of the radio.

2. Fill in the form. SN, Radio type, Location and Condition are required; the other fields are Radio ID, Asset owner, User / unit name, Department, Company, Document No. (PO / BA), Channel template, Last programming date, Last verification date and Notes.
3. Click **Save**. Changes of the holder, location, condition or programming are written to the history of the radio automatically, and the result appears at the bottom of the screen.

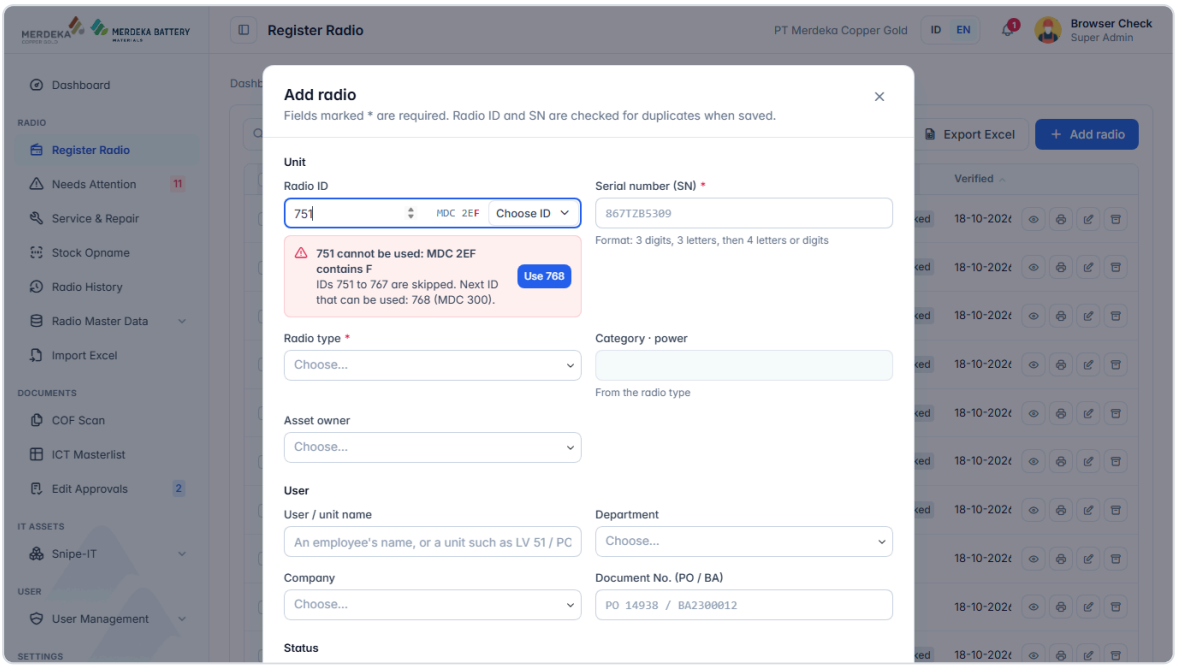


Figure 8. The Add radio form: Radio ID 751 is refused because its MDC 2EF contains F, and ID 768 is offered.

As soon as a Radio ID is typed, the form says whether it can be used:

- If it can, the panel **IDS AROUND IT** shows the IDs around it: free, in use, or containing F. Click a free ID to use it.
- If it cannot, the form gives the reason: the MDC Code contains F, another radio uses it (with its holder), or it is kept for a radio whose ID is empty for now. A button such as **Use 768** fills in the nearest replacement.
- The **Choose ID** button beside the field shows the next free IDs.

Other rules checked when saving:

- The **SN** is unique. The MDC Code, category and power are worked out automatically, and the **Owner** follows the asset owner.
- If somebody else saves the same radio after your form was opened, your save is refused with a message that the radio was changed just now: close the form, open it again, and make your changes again.
- While an Import Excel is running, saving is answered with "The register is busy"; try again a moment later.

Archiving, restoring and deleting

A radio that leaves for good is archived, not deleted. Click **Archive**, choose a reason from the cards (Moved to another site, Written off, Lost (BA signed) or Other), and fill in the date and a note. Its Radio ID is released so that another radio can use it. In the **Archived** view, **Restore** makes the radio active again; its old Radio ID comes back if no other radio uses it yet.

Deleting is only for data entered by mistake: the delete button is on the details page of the radio, and its confirmation asks for the Radio ID to be typed again.

Many radios at once

Tick the box at the start of a row to select a radio; the selection stays when you change pages or filters. The box in the column heading ticks one page, then **Select all N radios of the filter** selects every result of the filter. **Clear the selection** removes them all. The bar above the table only shows the buttons your role may use.

Button	What it does	Limit
Verify today	Writes a "Verification" history line on the chosen date	1,000 radios at once; archived radios are skipped
Archive	One reason, date and note for every ticked radio	100 radios ticked one by one
Delete	Only for data entered by mistake; confirmed by typing the number of radios	100 radios; cannot be undone in the application
Export selected	An Excel file with only the ticked radios	–

Radios that cannot be handled are named with their reason, and the others are handled anyway.

> To count the radios of one department in person, use **Stock Opname** on a phone (chapter 8). **Verify today** fits when the radios have already been checked some other way.

5. Radio details, handover and the checkout form

The details page shows one radio in full with its history. From here the radio is handed over to a new holder, and its checkout form is printed.

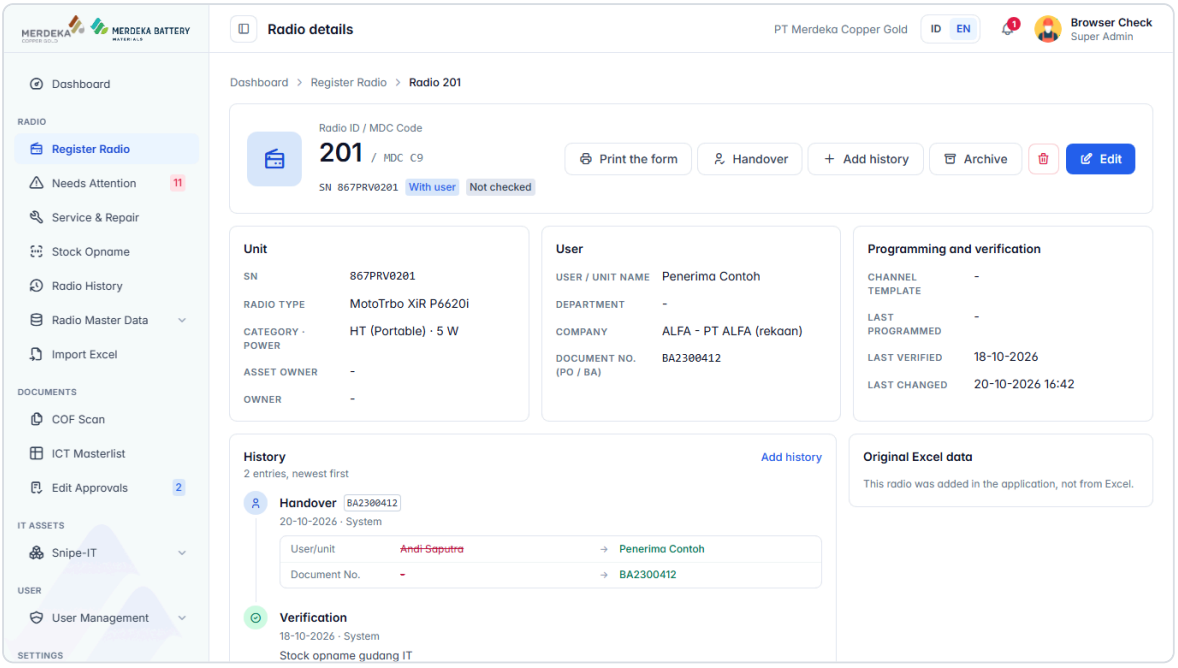


Figure 9. Radio details: the breadcrumb at the top, the buttons in the header, three cards of data, and the history with old-to-new changes.

What the details page holds

- **Header**: Radio ID, MDC Code, SN, location and condition, then the buttons **Print the form**, **Handover**, **Add history**, **Archive** (or **Restore**), delete, and **Edit**, as your role allows.
- **Check results**: the problems found on this radio, if any.
- **Unit**, **User**, **Programming and verification**: the data of the radio, including who changed it last.
- **History**: every event, newest first, with old values struck through and the new values beside them. **Add history** records an event by hand: Handover, Service, Programming, Verification or Note.
- **Original Excel data**: the row of the old register, for radios that came from Import Excel.

Handover

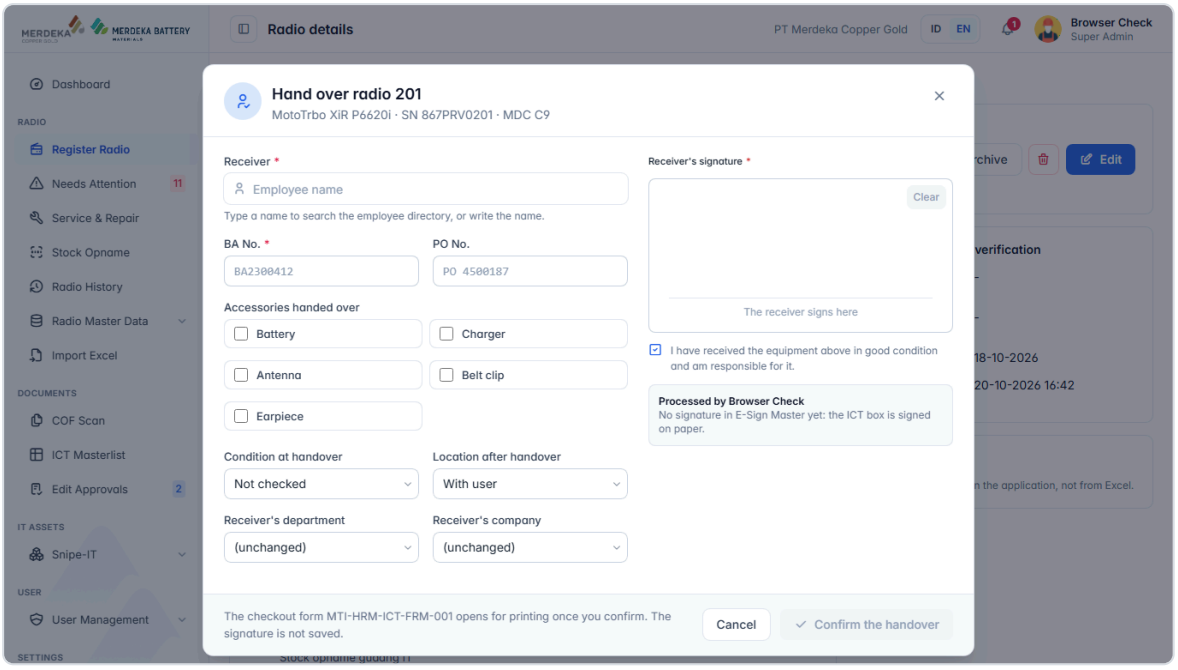


Figure 10. The handover dialog: receiver, BA No., accessories, condition, and the receiver's signature on the screen.

1. On the details page of the radio, click **Handover**.
2. Fill in **Receiver**: type a name to search the employee directory, or write the name if it is not found.
3. Fill in the **BA No.** (required) and the PO No. if there is one, tick the **Accessories handed over** (Battery, Charger, Antenna, Belt clip, Earpiece), then choose the **Condition at handover** and the **Location after handover**. The receiver's department and company may stay "(unchanged)".
4. Ask the receiver to sign in the box **Receiver's signature** with a pen, a finger or a mouse, and to tick the statement of receipt. **Clear** empties the box.
5. Click **Confirm the handover**. The button only becomes active once there is a signature.

Once confirmed, the radio belongs to the receiver, a "Handover" history line is written with the BA No. as its document number, and the checkout form opens for printing with the receiver's signature in the RESPONSIBLE USER box. The signature is stored nowhere, not in the audit either: it only exists on the printout.

The checkout form

The **Print the form** button (or the printer icon in Register Radio) opens the IT Equipment Checkout Form (MTI-HRM-ICT-FRM-001 Rev.000) for that radio, already filled in from the register and fitting one A4 page. Everybody who may see Register Radio may print it.

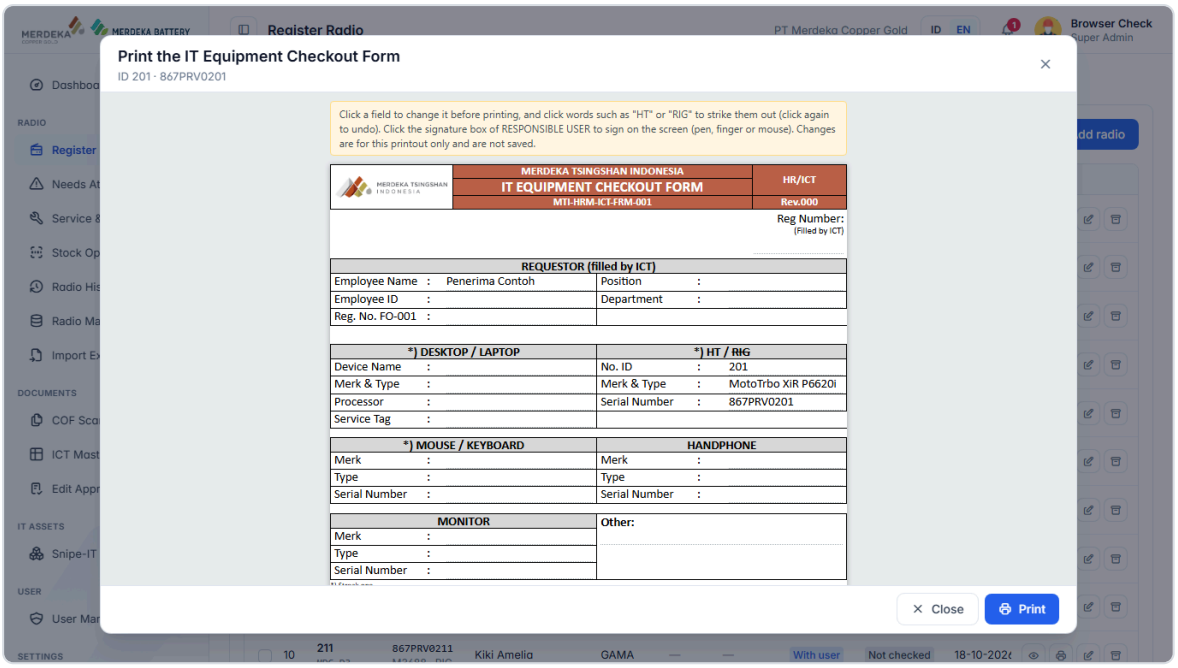


Figure 11. Print the IT Equipment Checkout Form: the form already filled in from the register.

1. Check what was filled in: No. ID, Merk & Type, Serial Number, Employee Name (REQUESTOR and RESPONSIBLE USER), Department, and the Date under PROCESSED BY. The choice HT or RIG that is not the kind of the radio is already struck out; click the word to strike it out or undo that.
2. Complete the other fields right in the dialog. Changes there are for that printout only and are not saved to the register.
3. If the receiver has not signed through a handover, click the signature box in the RESPONSIBLE USER block to sign on the screen, then **Use signature**. With a Windows Ink pen, the pressure sets the width of the line and the touch of a palm is ignored.
4. Click **Print** or press Ctrl+P, then choose a printer or "Save as PDF".
5. Close the dialog with the **Close** button. A click outside the dialog or the Esc key does not close it, so that nothing you typed is lost.

PROCESSED BY: if your account has a signature in **E-Sign Master**, your signature and name appear automatically, and the use is recorded. It is always the signature of the account that opened the form, never somebody else's. An account without a signature gets an empty box to sign on paper.

If what was filled in makes the form longer than one A4 page, a red warning shows on the screen, and the printout is scaled down automatically to stay on one page. The browser's header and footer are not printed.

6. Needs Attention

Needs Attention turns the results of the Checks column of Register Radio into a list of work: every problem becomes an item that is fixed, dismissed with a reason, or assigned to someone. The red number in the sidebar counts the open items.

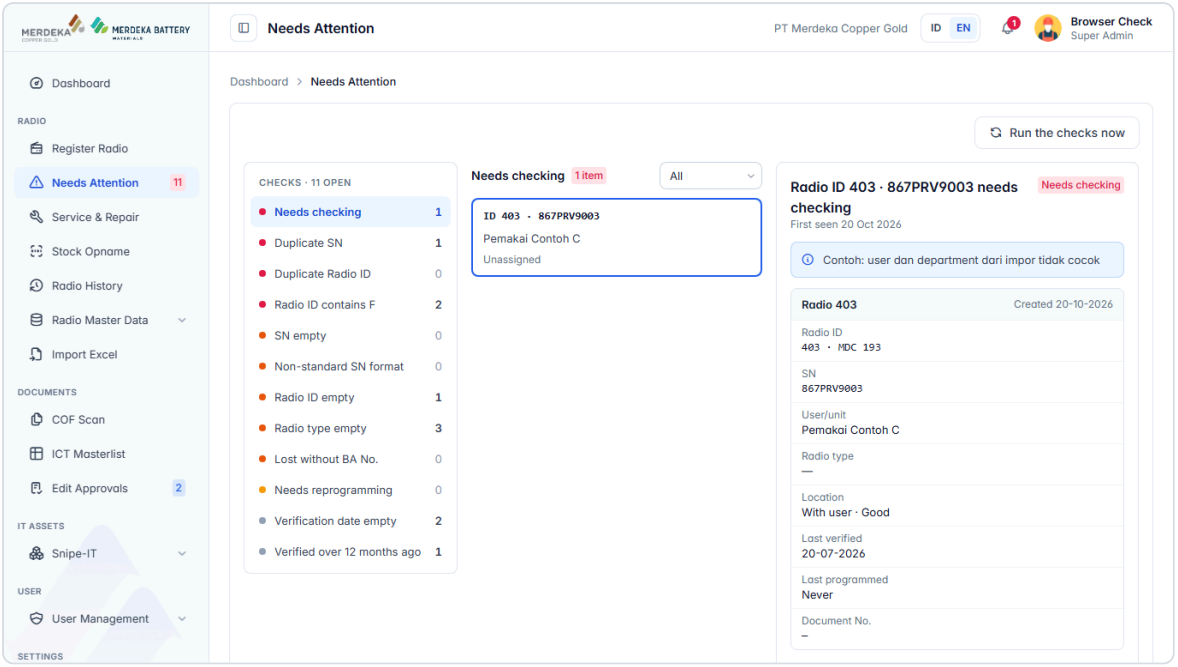


Figure 12. Needs Attention: the checks on the left, the items of the chosen check in the middle, the details of an item on the right.

Reading the page

- **Left**: every kind of check with the number of its open items. Choose one to see its items.
- **Middle**: the items of that check. One item per radio; for Duplicate SN and Duplicate Radio ID, one item per SN or ID that several radios use. Filter with **All**, **Assigned to me** or **Unassigned**.
- **Right**: the details of an item: when it was first seen, the data of its radio (for duplicates, the radios side by side with the columns that differ marked), the suggested fixes, and who it is assigned to.
- **Run the checks now** checks the whole register again at once.

Fixing

Choose a fix in the details of the item, fill in its value if asked, then click **Apply**. A fix is done like a change in Register Radio: the history and the audit are written.

Check	Fixes offered
Duplicate SN	Correct the SN of one radio, archive a radio as a duplicate, or "These radios really are different" (dismissed with a reason)
Duplicate Radio ID	Give one radio the next free ID, or archive it as a duplicate
Radio ID contains F	Use the next ID that can be used (the radio needs to be reprogrammed)
SN empty, Non-standard SN format	Fill in or correct the SN
Radio ID empty	Give back its last ID if it is still free
Radio type empty	Choose the radio type

Check	Fixes offered
Lost without BA No.	Fill in the BA No. of the loss, or fill in the BA No. and archive it as lost
Needs reprogramming	Record programming today
Verification date empty, Verified over 12 months ago	Record verification today
Needs checking	Mark as reviewed

Dismissing and assigning

- **Dismiss** closes the item with a reason, which is required. The item comes back when the data of its radio changes.
- Choose a name under **Assigned to** to give the item to an account that may change radio data. The item then shows under **Assigned to me** for that person.

Seeing this page needs the Read permission on the Needs Attention menu; assigning and dismissing need Update. Fixing needs Update on Needs Attention and on Register Radio.

7. Service & Repair

Service & Repair follows a radio with a problem from the report until it is back from the service center, so that you see how long radios stay there and what they cost.

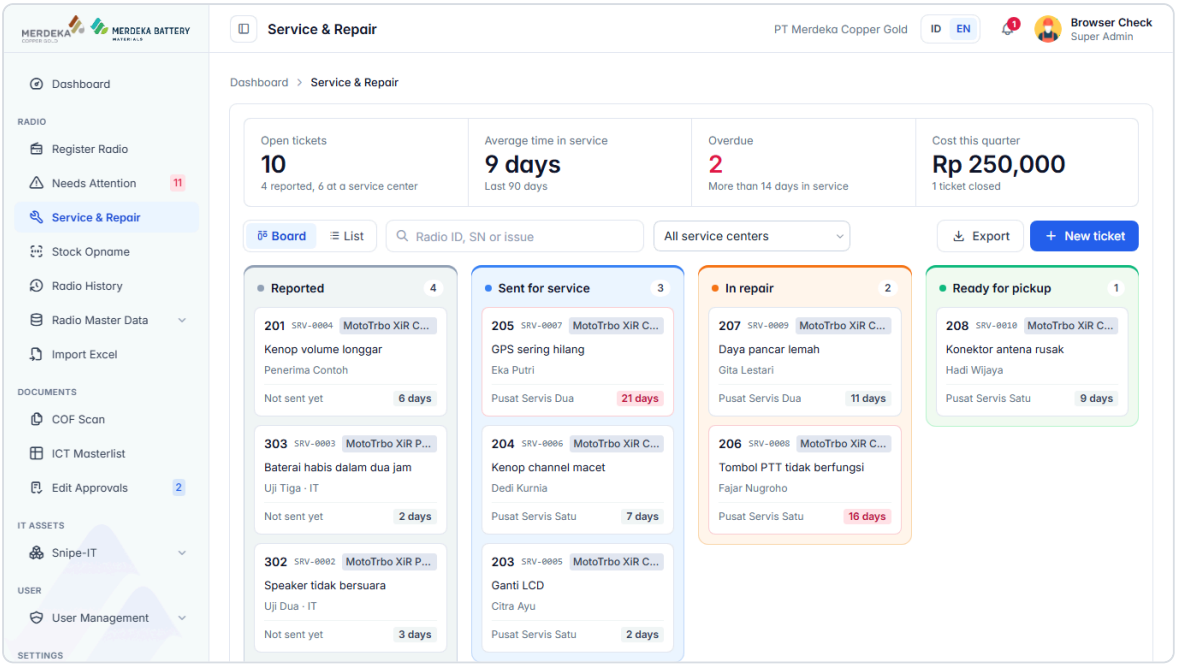


Figure 13. The Board view: one column per step, one card per ticket; a red number means overdue.

The steps of a ticket

Step	Button on the ticket	Fields	What happens to the radio
Reported	**New ticket** at the top of the page	Radio, Issue (required), Reported by	–
Sent for service	**Send for service**	Service center (required), Date sent, BA No., Expected back, Quote (Rp)	The location becomes In service; a "Service" history line is written
In repair	**Mark as in repair**	Quote (Rp), Expected back	–
Ready for pickup	**Mark as ready for pickup**	Cost (Rp)	–
Returned	**Mark as returned**	Date returned, Radio location afterwards (With user or IT stock, required), Condition when returned (required), Cost (Rp)	The location and condition are updated; a "Service" history line is written

The ticket number (for example SRV-0004) and the BA No. are written into the history of the radio. A radio can have only one open ticket, and an archived radio cannot get one.

This page

- **Four numbers** at the top: Open tickets, Average time in service over the last 90 days, Overdue tickets (more than 14 days in service), and the Cost this quarter.
- **Board** or **List**: cards per step, or a table. Search by Radio ID, SN or issue, and filter by service center.
- **Export** downloads an Excel file of all tickets, finished ones included.

- Click a ticket to open its details on the right: the steps, the facts of the ticket, the status of its radio (with the link **Open the radio**), and the activity. **Add a note** writes a note into the activity at any time.

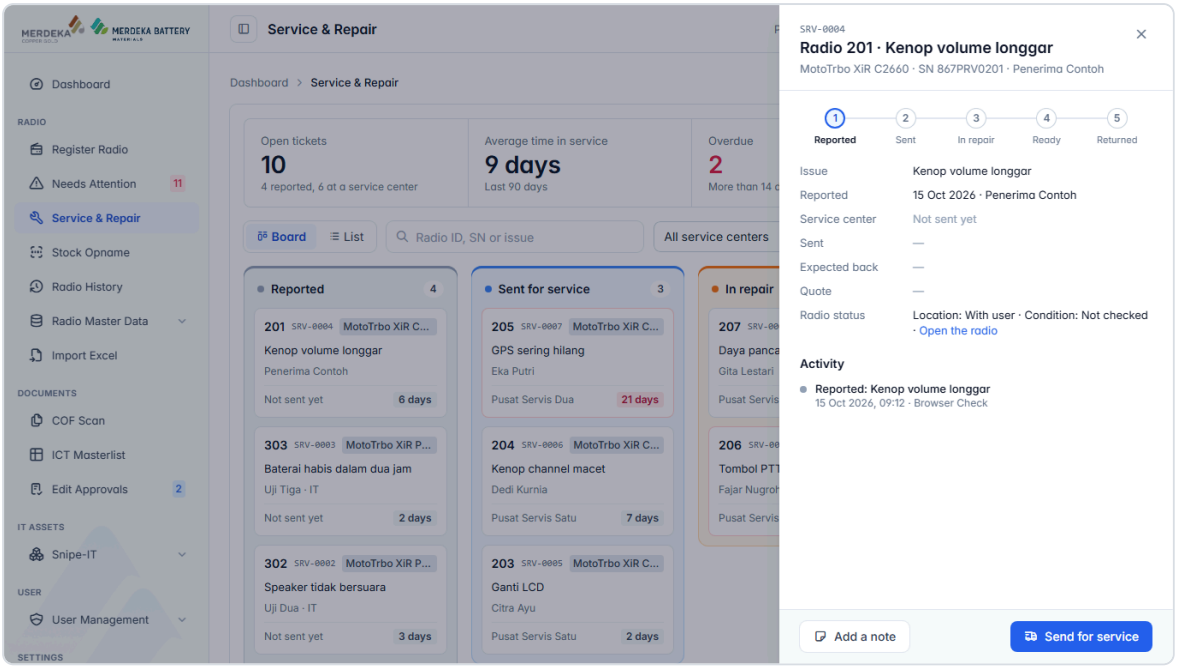


Figure 14. The details of a ticket: five steps, the facts of the ticket, the activity, and the button of the next step.

Creating a ticket needs the Create permission on the Service & Repair menu; moving it to the next step and adding notes need Update.

> While a ticket is still open, an Import Excel that deletes the old data is refused: those radios must come back first.

8. Stock Opname

Stock Opname is a page for phones: it counts the radios of one department by scanning the SN barcode, checks who holds each radio and its condition, and sends the result to the register. The page has no sidebar; the × at the top right goes back to the application.

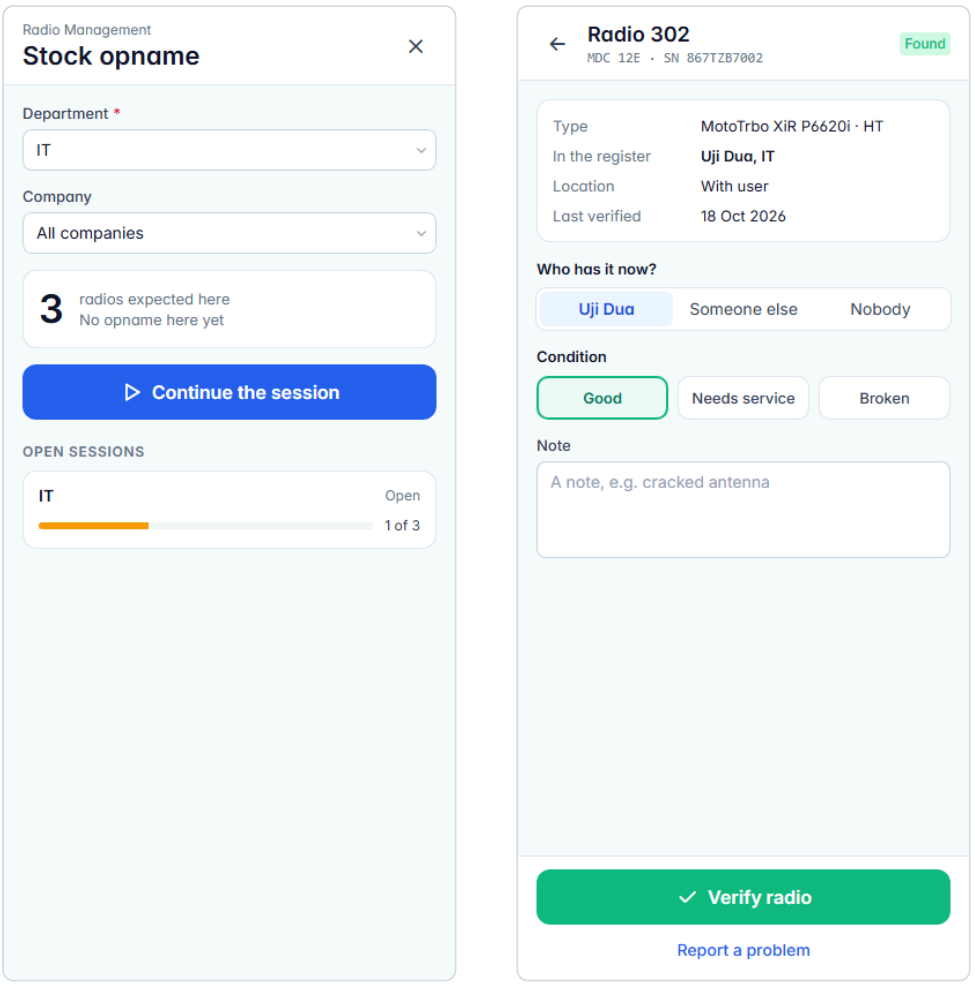


Figure 15. Left: choosing the department and continuing a session. Right: checking a radio that was found.

Steps

1. Open the **Stock Opname** menu on a phone. Choose the **Department** (and the **Company** if needed): the page says how many radios are expected there and when the last opname was.
2. Click **Start a session**. If a session for that department is still open, the button reads **Continue the session**; open sessions are also listed below it.
3. Scan the SN barcode on the radio with the camera. The camera works in Chrome on Android; in other browsers, or with a handheld scanner, type the Radio ID or SN and click **Find**.
4. For a radio that was found, answer **Who has it now?** (the holder in the register, **Someone else** with their name, or **Nobody**), choose the **Condition**, write a note if needed, then click **Verify radio**. Every scan is saved at once.
5. If the radio has a problem, **Report a problem** creates a ticket in Service & Repair; write the problem in the note first.
6. Click **See the summary**: how many are Verified, Not found, held by an Other holder, and from an Other department.
7. Click **Send to the register**. Every radio found today gets a "Verification" history line, and conditions that changed are updated.

Starting a session needs the Create permission on the Stock Opname menu; scanning and sending need Update. Reporting a problem also needs the Create permission on Service & Repair.

> While a stock opname session is still open, an Import Excel that deletes the old data is refused: send that session to the register first.

9. History, Master Data, E-Sign Master and Import Excel

Radio History

Radio History shows the events of all radios: handovers, service, changes of status, programming, verification and archiving, written automatically or added by hand. Filter with **Search** (SN, Radio ID, user, description, document number), **Kind** (several at once, with their counts), **From date** and **To date** (chosen in a calendar). The list follows the filters at once. The history of one radio is on the details page of that radio.

Radio Master Data

Master data are the choices used in the radio form. Each kind of master data has its own tab; the add and edit form opens in a card beside the table.

Tab	Fields	What it affects
Radio Types	Type name, Category, Power, Active	The category and power of a radio follow its type
Companies	Code, Company name, Active	Used for the user's company and the asset owner; an asset owner with the code MTI makes the Owner MTI
Departments	Department name, Active	The department of the holder of a radio, and the choices of Stock Opname
Channel Templates	Template name, Channels, Effective from, Active	A radio last programmed before the Effective from date is marked "needs reprogramming"

Master data that is made inactive can no longer be chosen for other radios; radios that already use it do not change. Master data can only be deleted when no radio uses it.

E-Sign Master

E-Sign Master keeps the signatures of the ICT staff for the PROCESSED BY box of the checkout form. By default only Super Admin can open it. It can be shown as a **Grid** (cards with a preview of the signature) or a **List**; the choice is remembered.

1. Open **Radio Master Data** → **E-Sign Master**, then click **Register a signature**.
2. Choose the account of the ICT staff member. One account has one signature.
3. The owner of the signature signs in the box with a pen, a finger or a mouse (or uploads a photo of the signature), then clicks **Save**.

A signature can be replaced, made inactive or deleted, and its **Use history** shows every use. It is a picture of a signature stored encrypted, not a certified digital signature.

Import Excel

Import Excel moves the old register (a file such as Radio Register 07-08-25.xlsx, the sheet Form and, when ticked, "to Konawe") into the application.

1. Choose the .xlsx file, then click **Start the import**.
2. Read the result in the number tiles. Every row with an SN or a user name becomes one radio; rows that only hold a number and a Radio ID are skipped.
3. Check the radios marked **Needs checking** through the link **Open in Register Radio**: their data contradicts itself in Excel (for example a duplicate SN) and was not guessed at.

Warning: when the register already holds radios, an import is only possible with the red box "Delete every radio and all history before the import" ticked, and the application asks for a confirmation once more. That choice deletes every radio and all history in the application, so do not use it on a register in use. Such an import is also refused while a service ticket or a stock opname session is still open.

10. COF Scan and ICT Masterlist

Both menus show what is in SharePoint inside the application: **COF Scan** the folder of scanned IT Equipment Checkout Forms, and **ICT Masterlist** the ICT Masterlist Excel workbook. The application reads them, so whoever is given these menus sees their content even without SharePoint access to that folder or workbook.

COF Scan

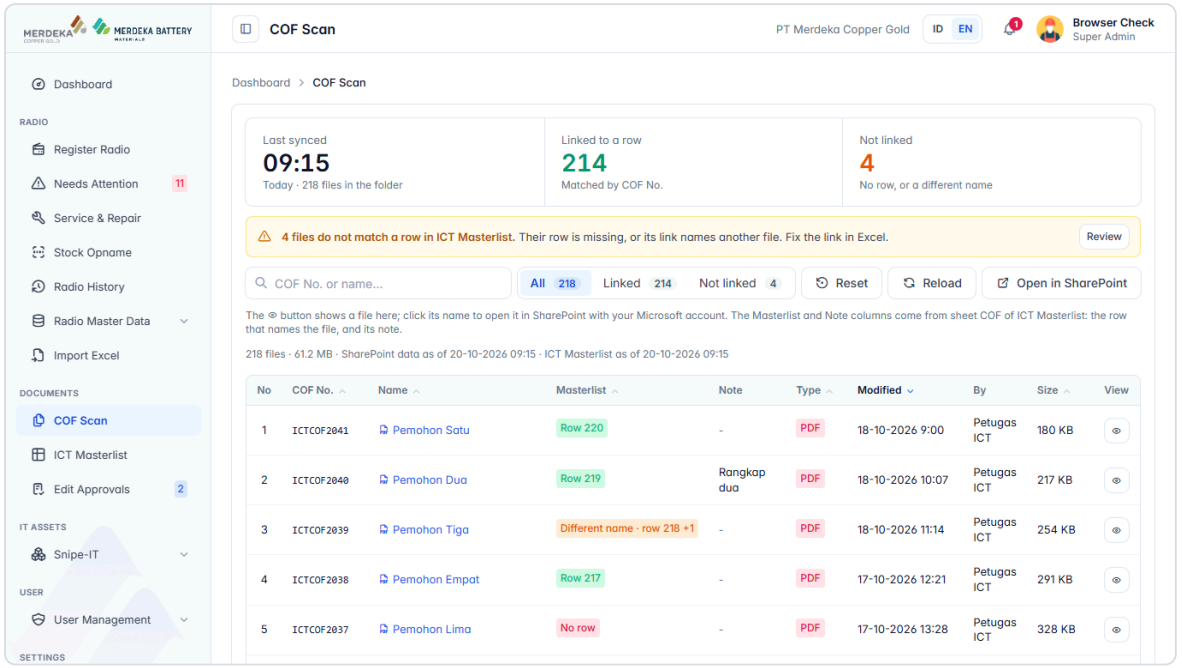


Figure 16. COF Scan: three numbers, the warning about files that do not match, and the filter All / Linked / Not linked.

- If you may also open the ICT Masterlist, three numbers at the top give **Last synced** (when the list was fetched and the number of files in the folder), **Linked to a row** (files that have their row in the ICT Masterlist), and **Not linked** (no row, or the row names another file). These numbers are for the whole folder, whatever is searched.
- When files are not linked, a yellow warning gives their number; **Review** shows only those files. Fix their links in Excel.
- **All**, **Linked** and **Not linked** filter the list by status.
- Names in the form `ICTCOF123 - Name.pdf` are split into the columns **COF No.** and **Name**. Search by number or name, and click a column heading to sort.
- The **Masterlist** column gives the Excel row of each file ("Row N", "Different name · row N" or "No row") with a link to that row, and the **Note** column shows the Note column of Excel.
- The eye (**View**) shows a PDF or picture of up to 10 MB in a dialog, with previous and next buttons. Click the name of a file to open it in SharePoint with your Microsoft account.
- A newly uploaded file shows within 10 minutes; **Reload** fetches it at once. **Reset** clears the search, the filters and the sorting.

ICT Masterlist

- Every sheet that is visible in Excel becomes a tab, with its number of rows. The **Row** column is the row number in Excel; the newest rows come first.
- Cells show as in Excel: dates as dd-mm-yyyy, and formulas as their result when the workbook was last saved.

- The search looks in every column of every sheet at once; each tab shows how many of its rows match. When several words are typed, a row must hold all of them.
- The banner above the table says how you may change the workbook (chapter 11). **Open in Excel** opens the workbook in Excel.

Cross-checking in the COF tab

In the COF tab of the ICT Masterlist, the **File scan** column and five cards above the table (Total rows, Found, Missing, Different name, Subfolder) show whether each row has its scanned file in the COF Scan folder. Click a card to see only its rows; click it again to see them all.

Status	Meaning	What to do
Found	The file named by the link of the row is in the folder	–
Different name	No file has the name in the link, but a file has the same COF number	Fix the link in Excel; even a double space makes a name different
Subfolder	The link points into a subfolder, usually because the requester's name contains a "/"	Fix the link in Excel
Missing	No file has that name or number	Make sure the scan has been uploaded to the folder

11. Changing the ICT Masterlist

The ICT Masterlist can be changed from the application in two ways. Those who may only ask for changes (for example Admin Radio) ask for a change of one cell, and somebody else approves it. Those who may approve (Document Controller and Super Admin) change a row directly.

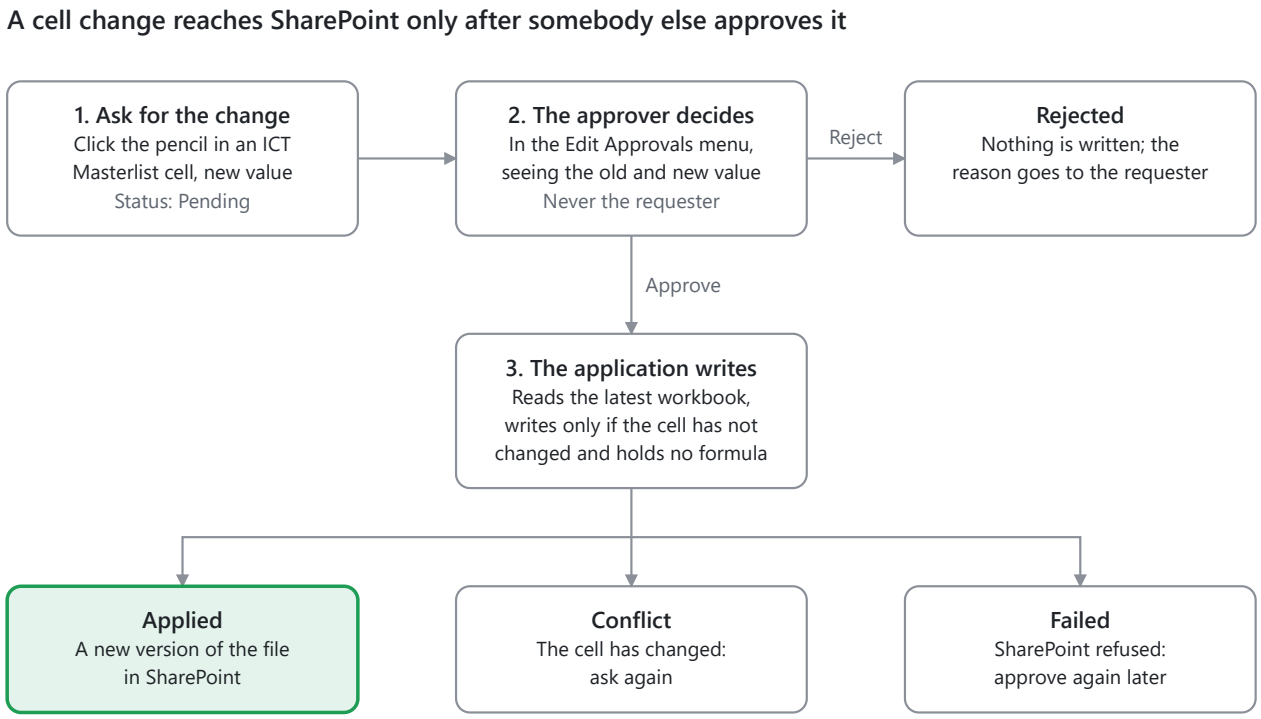


Figure 17. How a request to change a cell goes: two people, four possible results.

A cell is only written if it still holds what it held when the change was asked for, so a change made by somebody else in Excel is never overwritten.

Asking for a change

1. Open **ICT Masterlist** and find the row.
2. Click the pencil in the cell to change. A cell with a formula carries a padlock and can only be changed in Excel; the File scan column cannot be changed either.
3. Fill in the new value. Write numbers as 1500 or 1.500,50; dates as 15-03-2023 or 15-03-2023 14:30; anything else is kept as text, as it is (000123 stays 000123). Leave it empty to empty the cell.
4. Click **Ask for the change**. The cell is marked "waiting" until the request is decided; a cell can have only one request waiting.

Approving or rejecting

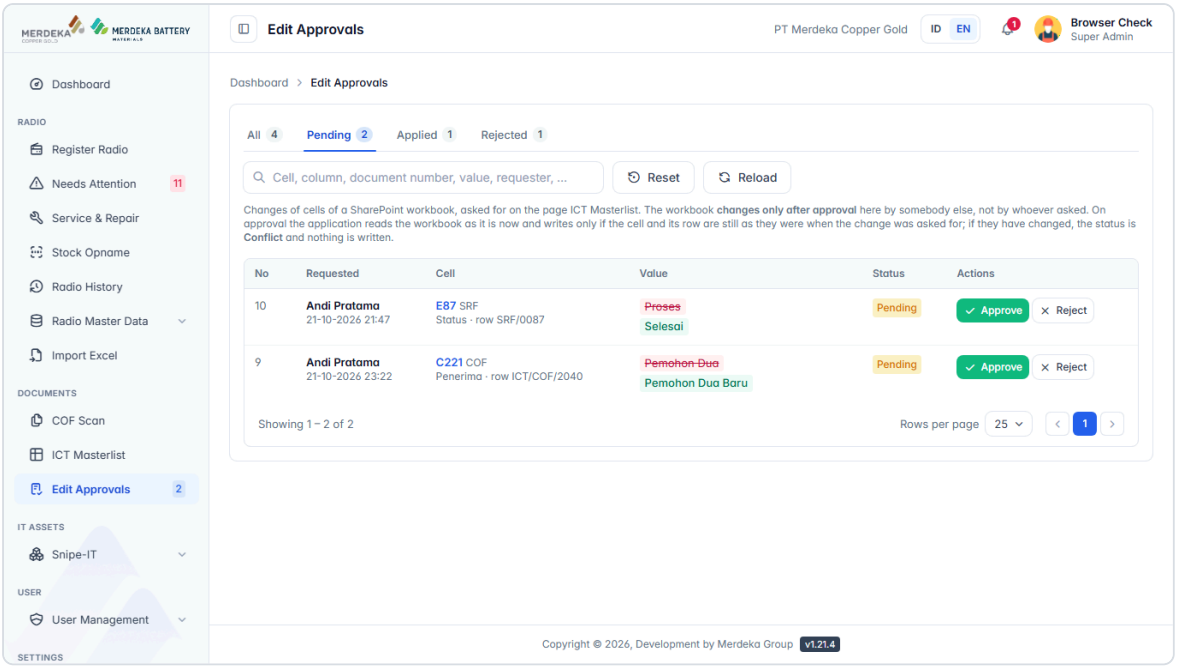


Figure 18. Edit Approvals: a tab per status, old values struck through in red and new values in green.

1. Open **Edit Approvals**. A notification from the bell opens the request it is about.
2. Choose a status tab (All, Pending, Applied, Rejected; Processing, Conflict and Failed appear when there are such requests), then compare the old and the new value.
3. Click **Approve** for the application to write the cell to SharePoint, or **Reject** with a reason.

Nobody can approve their own request, not even Super Admin. An approver who rejects their own request withdraws it.

Status	What to do next
Pending	Wait for the decision of an approver
Applied	Nothing; the previous version can still be restored from the version history in SharePoint
Rejected	Ask again if it is really needed
Conflict	Reload the page, then ask again from the latest value
Failed	Approve again later, for example once the file is closed in Excel on a desktop

Editing a row directly

For those who may both ask for and approve changes (in practice the Document Controller and Super Admin), the pencil is on the row number instead of in every cell, and their changes are written at once, without anybody else's approval.

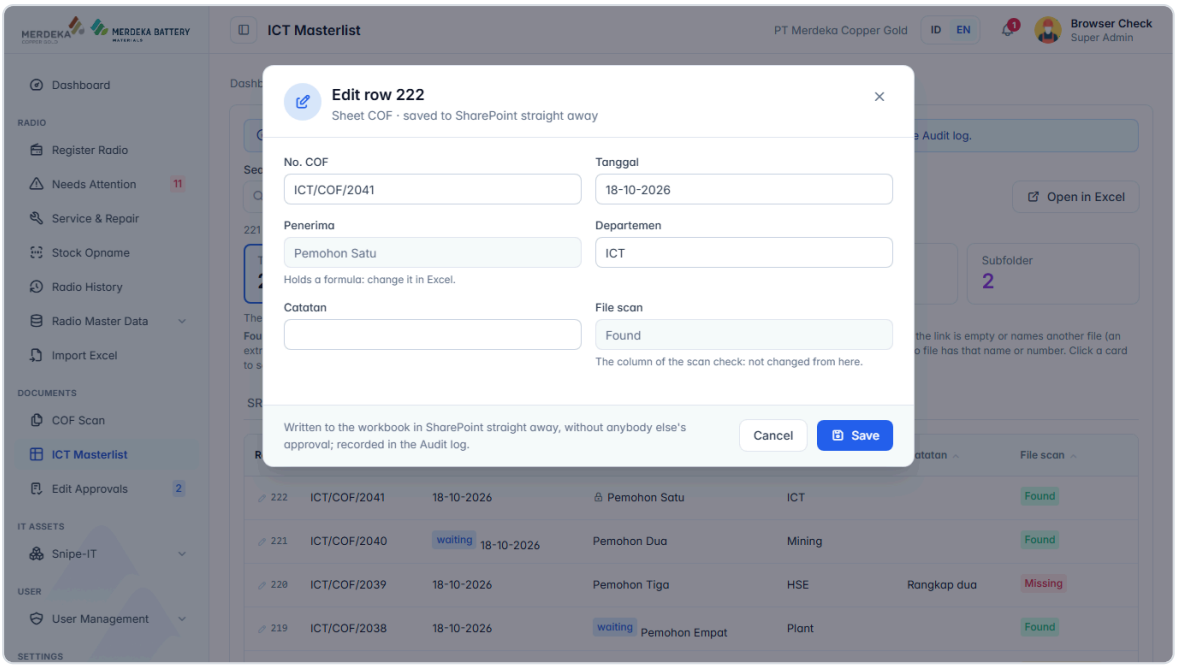


Figure 19. Edit row: every column of the sheet; formula cells and the File scan column are read-only.

1. Click the pencil on the row number. The **Edit row** dialog holds every column of that sheet.
2. Change what is needed. A cell with a formula, a cell with a request waiting, and the File scan column are read-only; the reason is written under the field.
3. Click **Save**. Only the fields that changed are sent, and the row is written to the workbook in one save.

Nothing is written if the row has changed in SharePoint since the page was loaded, if somebody else saved the workbook in between, or if the workbook is open in Excel; the message says which. Every cell written is recorded as a request applied by its own author ("Written directly by ..." in Edit Approvals) and in the Audit Log.

Only the content of cells can be changed; adding or deleting rows and columns is still done in Excel. Formulas that depend on the cell are only recalculated when the workbook is opened in Excel.

12. IT Assets: Snipe-IT

The **Snipe-IT** menu in the IT ASSETS section shows what Snipe-IT, the company's inventory of IT assets, holds, without opening Snipe-IT itself. The menu only reads: nothing can be changed from here, and the application only reads from Snipe-IT.

By default only Super Admin gets this menu. Other roles can be given it in **Role & Permission**: the Snipe-IT group and the pages they may open, with the Read permission.

> When Snipe-IT cannot be reached, these pages say so (chapter 14). Register Radio and the other menus keep working as usual.

Overview

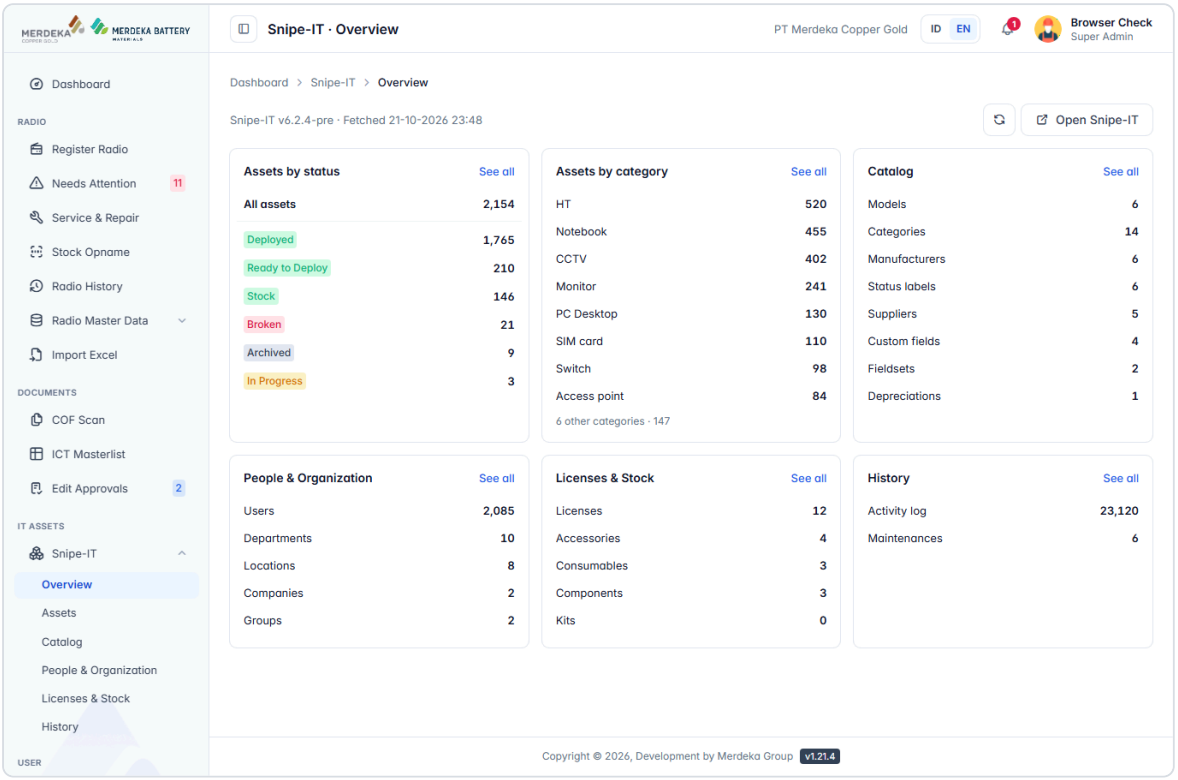


Figure 20. Overview: assets by status and by category, then the size of every list in Snipe-IT.

- The top line gives the version of Snipe-IT and when its data was fetched.
- **Assets by status**: the number of all assets, then per status (for example Deployed, Ready to Deploy, Broken). **Assets by category**: the categories with the most assets, the rest grouped ("6 other categories").
- The panels **Catalog**, **People & Organization**, **Licenses & Stock** and **History** give the size of each list. **See all** opens its page.
- The data of Snipe-IT is kept for a short while (5 minutes by default) so that moving between pages stays fast. The circular-arrow button (**Fetch again from Snipe-IT**) fetches it again at once, and **Open Snipe-IT** opens Snipe-IT in a new tab.

Lists

Each of the other pages holds one or more lists of Snipe-IT, one tab per list:

Page	Lists
Assets	All assets
Catalog	Models, Categories, Manufacturers, Status labels, Suppliers, Custom fields, Fieldsets, Depreciations
People & Organization	Users, Departments, Locations, Companies, Groups
Licenses & Stock	Licenses, Accessories, Consumables, Components, Kits
History	Activity log (checkouts, checkins, changes), Maintenances

- Type in the search box to search that list, click a column heading to sort, and move between pages below the table.
- The Assets list can be filtered by category and by status (with their counts), and the Activity log by action (Checkout, Checkin, Created, Updated, Deleted, Audited). The **Changes** column of the Activity log uses readable field names and values, for example "Status: Ready to Deploy → Deployed".
- Names, statuses and notes show as they are written in Snipe-IT, without translation.

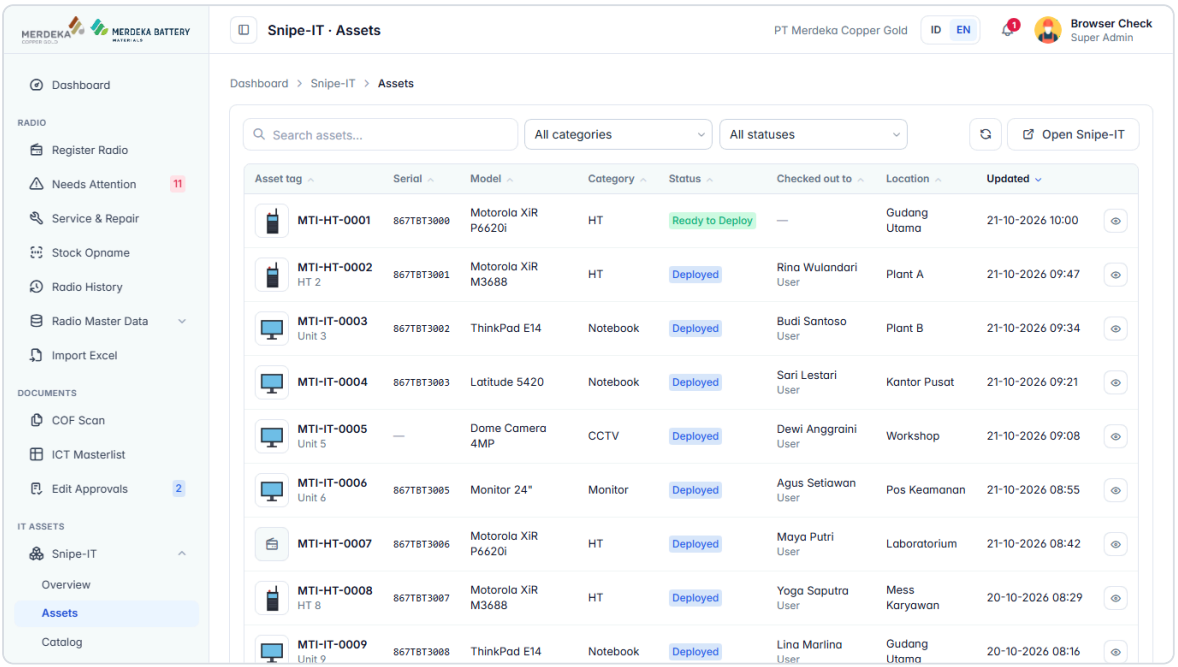


Figure 21. The Assets list: the picture, asset tag and name of each asset, the category and status filters, and the eye button at the end of each row.

In the Assets list, each row starts with the picture of the asset (the picture of the asset or of its model in Snipe-IT; without one, the icon of its category), with the asset tag and the name of the asset below it. The eye button (**See the details**) at the end of the row opens the details of the asset in this application. An asset named in the Activity log or Maintenances opens its details too.

Asset details

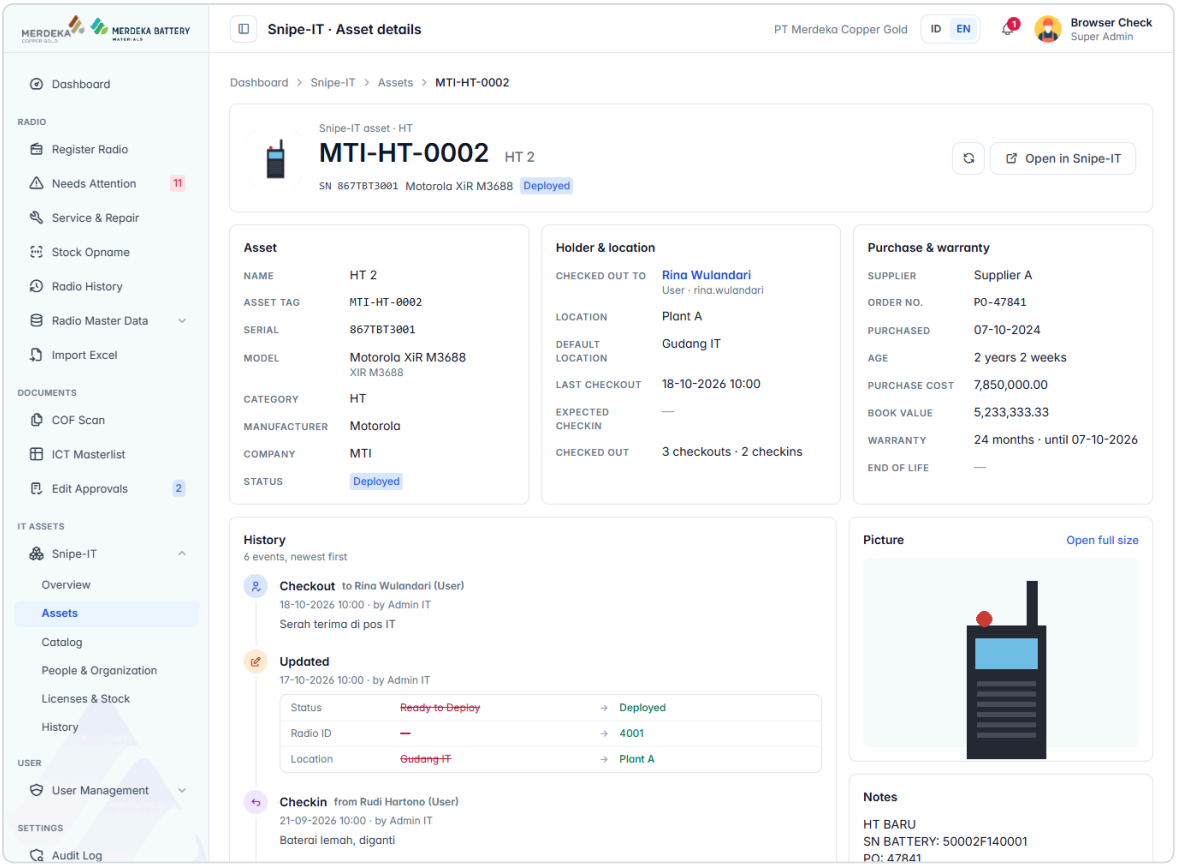


Figure 22. Asset details: the data of the asset, holder and location, purchase and warranty, history, and the picture.

Panel	What it shows
Header	Asset tag, name, serial, model and status; the buttons **Fetch again from Snipe-IT** and **Open in Snipe-IT**
Asset	Name, asset tag, serial, model, category, manufacturer, company, status
Holder & location	Checked out to, location, default location, last checkout, expected checkin, and how many times the asset was checked out and in
Purchase & warranty	Supplier, order No., purchase date, age, purchase cost, book value, warranty (and until when), end of life
History	The events in Snipe-IT, newest first: checkouts, checkins, and changes with the old value struck through and the new one beside it. When there are more than 100, only the 100 newest show; the rest is in Snipe-IT
Picture	The picture of the asset or its model; **Open full size** opens it at its real size
Notes, Custom fields	The notes of the asset, and the custom fields of its model (for example Radio ID and MDC Code)
Maintenances, Licenses, Assets attached	The maintenance of this asset, the licenses checked out to it, and other assets checked out to this asset
Dates & audit	When the asset was created and updated, the last and the next audit

Custom fields whose name marks a secret (for example a password, PIN, BitLocker key or other key) read **Hidden**, and so do password values in the history; license keys are never sent to the browser. The components and files of an asset are not available through the API of Snipe-IT: see them in Snipe-IT with the **Open in Snipe-IT** button.

13. Administration

The menus of the USER and SETTINGS sections are used by Super Admin to manage accounts and roles, read the audit trail, and create API tokens.

Adding an account

1. Open **User Management** → **Registered Users**, then click **Add user**.
2. Choose the **Sign-in method**: **Microsoft** (signs in with the Microsoft button) or **LDAP** (signs in with the Windows password in the Merdeka Account form; the Microsoft button still works).
3. Type a name or e-mail address in **Find employee**, then choose the person. For LDAP the search looks in Active Directory and fills in Name, Email and Directory username; for Microsoft it looks in the HR database and fills in Name and Email.
4. Choose the **Role** and the **Status**, then click **Submit** and answer the confirmation "Is the data correct?". For LDAP, the Directory username is checked against Active Directory first; a username that does not exist there or is disabled is refused.

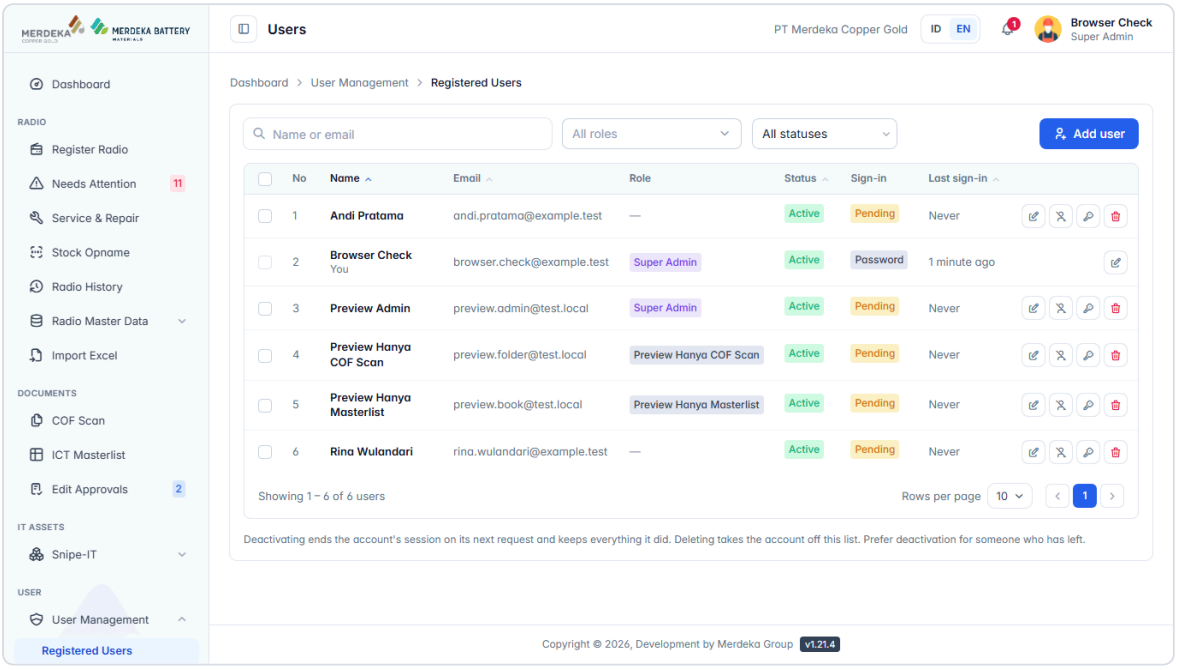


Figure 23. The Registered Users page: the accounts with their role, status, sign-in method and last sign-in.

The list of users shows No, Name, Email, Role, Status, Sign-in and Last sign-in; filter by name or e-mail, role and status. The **Sign-in** column shows how an account signs in:

Sign-in	Meaning
Microsoft	Signs in through Microsoft
LDAP	Has a directory username and has not signed in through Microsoft yet
Both	Has a directory username and has signed in through Microsoft
Password	Signs in with a password made by an admin
Pending	Has never signed in; if it stays like this, compare its e-mail address with the person's Microsoft address

The buttons of a row: Edit (pencil), Deactivate or Activate, Set password (key; not for LDAP accounts, whose password is in Active Directory), and Delete. A deactivated account is refused at sign-in, and its open session ends on its next

click. For many accounts at once, tick their rows and use **Activate**, **Deactivate** or **Delete** in the bar above the table (at most 100 accounts; Delete is confirmed by typing their number). Your own account cannot be deactivated or deleted.

Role & Permission

Click **Add Role**, fill in the name of the role and its **Home Page** (the first page after signing in), then tick the permissions per menu under **Menu & Permission**. The permissions are Read, Create, Update and Delete. What they mean on the radio menus:

Permission	Register Radio	Needs Attention	Service & Repair	Stock Opname
Read	List, details, export, print the form	See the inbox	See tickets, export	Open the page
Create	Add radios	–	New tickets	Start a session
Update	Edit, hand over, archive and restore, add history	Assign, dismiss; fix (also needs Update on Register Radio)	Move to the next step, notes	Scan, send to the register
Delete	Delete radios	–	–	–

On Master Data, Read means seeing the lists, and Create, Update, Delete mean adding, editing and deleting. On Import Excel, Create means importing into a register that is still empty; an import that deletes the old data also needs Delete on Register Radio. On Snipe-IT only Read matters. The Super Admin role can only be given or taken away by a Super Admin.

Audit Log

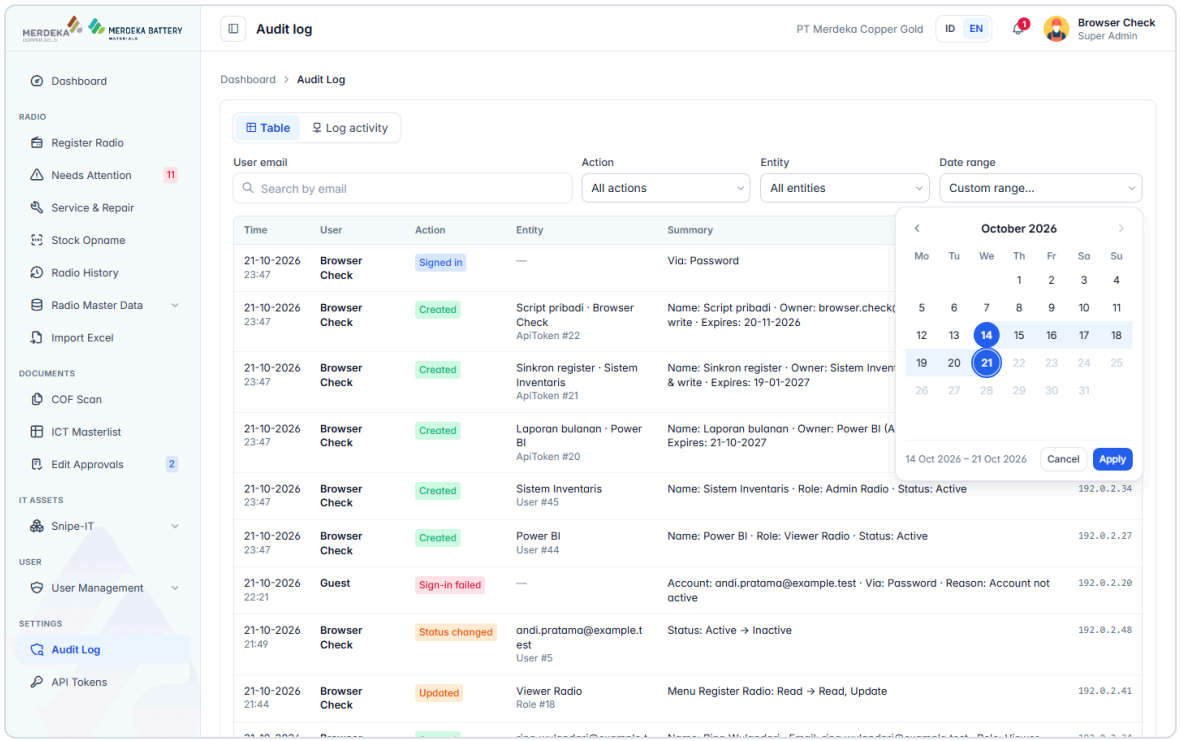


Figure 24. Audit Log: the filters User email, Action, Entity and Date range, with the calendar of Custom range open.

The Audit Log records who signed in (successfully, failed with the reason, signed out, session expired) and every change of data with its values before and after. Audit entries cannot be changed or deleted, and stay as they were written.

- The **Table** view shows Time, User, Action, Entity, Summary and IP; the **Log activity** view shows them as a timeline.

- Filter by **User email**, **Action**, **Entity** (the kind of record: Account, Radio, Radio type, and more) and **Date range**: Today, Yesterday, Last 7 days, Last 30 days, All time, or **Custom range…**. In its calendar, click the first and then the last day, then **Apply**; **Cancel** or Esc leaves it.
- The choices of **Action** have readable names, for example "Sign-in failed", "Account status changed" or "API token revoked", grouped by part of the application: Sign-in, Accounts, Roles, Radios, Service & Repair, Needs Attention, Master data, E-Sign, Documents and API.
- The address of the page follows the filters, so a view can be shared.
- Click a row to open the **Event details**: the time to the second, the user, action, entity, IP, the **Browser** (for example "Chrome 141 on Windows") and the changes line by line.

API Tokens

API Tokens are used when another application (for example Power BI) reads or changes the register through the API `/api/v1`; its documentation is at `/docs/api` (the **API documentation** button).

- An **API client** (made by Super Admin with **New client**) is the account of an application with one role. It cannot sign in on a screen and is not listed in User Management; making it inactive or deleting it stops all its tokens.
- A **personal token** acts as its owner with the owner's role, and stops when that account is deactivated.
- Every token is made with **New token**: **Read only** or **Read & write**, expiring in 30, 90, 180 or 365 days, and it can be revoked at any time (**Revoke**). A token is shown only once, when it is made: copy and keep it then.
- The owner of a token gets a notification 7 days before it expires.

14. Messages you may see

Messages appear in the chosen language; this table gives their English text. If a message is not in this table, note its text and the version in the footer, then contact the admin of the application.

Message	Meaning	What to do
Wrong username or password!	The e-mail address, username or password is wrong	Check what you typed; an LDAP account uses the Windows password
Your account is not registered!	The application has no account for you yet	Ask an admin to add your account
The user is not registered in Microsoft SSO!	The e-mail address of your Microsoft account is not registered in the application	Ask an admin to register exactly the e-mail address of your Microsoft account
No access yet	Your account has no role yet	Ask an admin to give it a role
Account inactive	Your account has been deactivated	Ask an admin to activate it
You are signing in with a public e-mail address!	Signing in with a private e-mail address	Use your work Microsoft account
The directory cannot be reached	Active Directory cannot be reached right now	Try again shortly, or sign in with Microsoft
Microsoft sign-in was not completed, Sign-in session expired	The Microsoft sign-in was cancelled or took too long	Click **Microsoft** again
Too Many Requests	More than 10 sign-in attempts within a minute	Wait a minute
Session expired	The session ended: 30 minutes without activity, or 8 hours after signing in	Sign in again
Your session has ended.	The session ended while a page was still open	Reload the page, then sign in again
You do not have access to this page	Your role has no permission for that menu	Click **To the home page**; ask an admin if you really need the access
This radio was changed just now by ...	Somebody else saved the same radio after your form was opened	Close the form, open it again, then make your changes again
The register is busy	An Import Excel is running	Try again a moment later
... cannot be used: MDC ... contains F	The MDC Code of that ID contains the letter F and cannot be used	Use the replacement ID the form offers
Someone else has moved this ticket on. Reload the page.	Somebody else moved the same ticket to its next step first	Reload, then check its step
No radio in the register has that Radio ID or SN.	The radio scanned in Stock Opname is not in the register	Check the barcode or what you typed; register the radio if it is really new
N service tickets are still open / N stock opname sessions are still open	An Import Excel that deletes the old data is refused	Finish the tickets or send the opname session to the register first
Row N has changed in SharePoint since the page was loaded	Editing a row directly: the content of that row changed after the page was opened; nothing was written	Reload the page, then try again

Message	Meaning	What to do
Somebody else saved the workbook in SharePoint after it was read	Editing a row directly: the workbook was saved in between; nothing was written	Reload the page, then try again
The workbook is locked because somebody has it open in Excel	The workbook is open in Excel on a desktop; nothing was written	Try again in a moment
Snipe-IT cannot be reached. Try again in a moment.	The Snipe-IT server is not answering right now	Try again a moment later; the other menus still work
Snipe-IT refused the token, Snipe-IT is not set up	The settings of Snipe-IT on the server are not right yet	Contact the admin of the application
This asset is not in Snipe-IT.	The asset was deleted in Snipe-IT, or the address is wrong	Go back to the Assets list and find the asset again